

WordMind — Semantic Memory Battle



An interactive word association game powered by natural
language processing

Student: MAHI PANCHAL JIYA VYAS

Enrollment: 202302626010095, 202302626010145

Div: B

THE PROBLEM

Word association is easy to play, hard to judge

Meaning can connect words that
spelling never will.



Rules are manual

Traditional word games often depend on manually defined rules or basic word matching.



Spelling misses meaning

“doctor” and “hospital” are not character matches, but they express related meaning.



WordMind measures the link

NLP makes the relationship semantic, measurable, and interactive.

PROJECT CONCEPT

The project turns **meaning** into a game mechanic



Browser-based word association game

For one human and AI opponents.



Represent word meaning

With pretrained sentence embeddings.



Measure relatedness

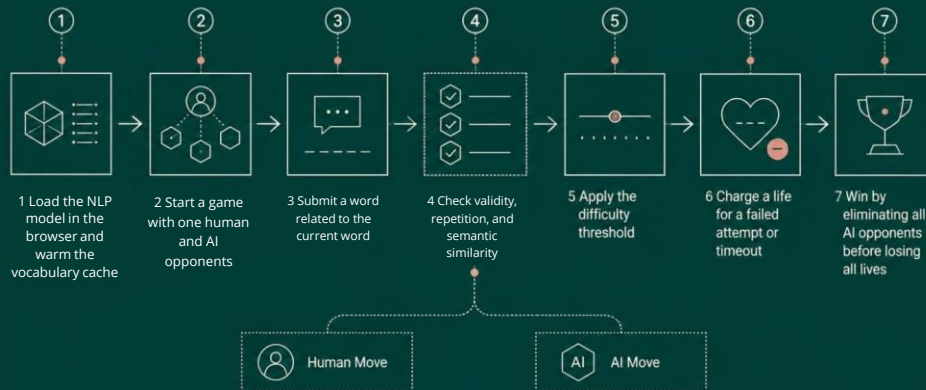
With cosine similarity.



Support

Difficulty, timers, lives, duplicate detection, hints, and an evaluation module.

EMBEDDING SPACE (PCA)



IMPLEMENTATION

The browser contains the **full NLP stack**



HTML + CSS

Interface, page structure, responsive visual design



JavaScript

Turns, timers, lives, game engine, AI decisions, and evaluation logic



Transformers.js

Loads and runs the NLP model in the browser



ONNX + WebAssembly/WebGPU

Browser inference were supported



Hugging Face CDN (all-MiniLM-L6)

Model download on initial load; no custom backend or API key for NLP inference

REPRESENTATION

Embeddings give each word a position in meaning-space

The pretrained Xenova/all-MiniLM-L6-v2 model turns words into numerical vectors.

384-dimensional embeddings · mean pooling · L2 normalization

The vector captures useful semantic relationships, but it does not understand words exactly like a human.

Example neighborhood

doctor → numerical vector

hospital → numerical vector

nurse → numerical vector

Semantic neighborhood

doctor · hospital · nurse

COMPARISON RULE

Cosine similarity compares direction, not spelling

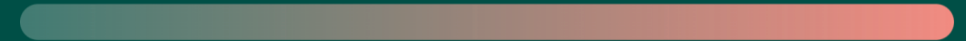
$$\cos(\theta) = \frac{A \cdot B}{\|A\| \times \|B\|}$$

A and B: embedding vectors for two words

Dot product: how strongly the vectors point together

Vector magnitudes: normalize scale so the comparison focuses on direction

Acceptance thresholds



0.35

Easy

0.42

Normal

0.50

Hard

0.20–

0.80

Custom

These are acceptance thresholds, not accuracy percentages.

Related words can produce stronger similarity than unrelated words, but the actual score depends on the model and pair.

The NLP Lab measures the **decision rule**

Evaluation tests the threshold, not a claimed experimental score.

Evaluation element	Implemented plan and reported result
Dataset	Manually labelled dataset: 65-word pairs — 43 related and 22 unrelated
Threshold sweep	Evaluate cosine thresholds from 0.20 to 0.70 in 0.05 increments
Metrics	Report accuracy, precision, recall, and F1-score
Baseline	Compare with character-level Jaccard similarity as an evaluation-only baseline
Output	Identify the best F1 threshold and export evaluation results as CSV

No experimental score is claimed here without measured results.